

RAG-Based Fake Review Detection System

Problem Statement

Online platforms depend on customer reviews for decision making. Fake or misleading reviews reduce trust and affect business reputation. This project proposes a Retrieval-Augmented Generation (RAG) based Fake Review Detection System that combines information retrieval and large language models to identify fake reviews with higher accuracy. The system retrieves relevant contextual information from a knowledge base and uses it to analyze review authenticity, provide explanations, and support better decision-making.

Objectives

1. Detect fake reviews using NLP and RAG techniques.
2. Retrieve supporting evidence from a knowledge base.
3. Improve classification accuracy and transparency.
4. Generate explainable outputs for users and administrators.